

Ph134, Lecture 4

- **Outputs of a χ^2 fit and how to interpret them.**
- **Discussion of fitting**

Output of a χ^2 / least squares / Gaussian max likelihood fit:

Fit parameters ($a, b, \dots$) that minimize

$$\chi^2 \equiv \sum_{m=1}^M \frac{[\bar{y}_m - f(x_m)]^2}{s_{\bar{y}_m}^2}$$

Parameter uncertainties ($s_a, s_b, \dots$) that tell what shifts in ($a, b, \dots$) would reduce Prob(our results) by a factor of $\sqrt{e}$

Final minimized value of χ^2 and of χ^2 per degree of freedom (also called reduced χ^2): $\# d.f. = M - \# \text{fit params}$

PTE or “probability to exceed”: if the functional form does describe the underlying relationship and the SEMs used were good estimates of error, what is the probability that random errors give you a χ^2 bigger than you got, in a fit with this many degrees of freedom.

Interpreting χ^2 value:

~~If $\chi^2 = \sum_{m=1}^M \left(\frac{\bar{y}_m - f^{true}(x_m)}{s_{\bar{y}_m}} \right)^2$, then $\left\langle \frac{\chi^2}{M} \right\rangle = 1$.~~

But $\chi^2 = \sum_{m=1}^M \left(\frac{\bar{y}_m - f^{fit}(x_m)}{s_{\bar{y}_m}} \right)^2$,

and $f^{fit}(x)$ has p parameters that the data points themselves helped to determine. Thus $\left\langle \frac{\chi^2}{M-p} \right\rangle = 1$.

Chi squared per degree of freedom (reduced chi squared):

$$\frac{\chi^2}{d.f.} = \frac{\chi^2}{M-p}$$

$$\left\langle \frac{\chi^2}{d.f.} \right\rangle = 1$$

Reduced chi squared is also denoted $\chi_v^2 = \frac{\chi^2}{v}$
where $v = M - p$ is the number of degrees of freedom.

Meaning of $\frac{\chi^2}{d.f.}$ (reduced chi squared):

$$\frac{\chi^2}{d.f.} \ll 1?$$

-this experiment was lucky?

-uncertainties of data were overestimated?

(if so, reported uncertainties of fit parameters are suspect)

-different data points were sampled in correlated ways?

$$\frac{\chi^2}{d.f.} \gg 1?$$

-this experiment was unlucky?

-uncertainties of data were underestimated?

(if so, reported uncertainties of fit parameters are suspect)

-the curve fit hasn't found the global minimum of χ^2 ?

-the function just doesn't capture reality?

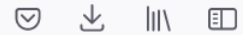
(wrong or incomplete theory? systematic error not found?)

What is $\ll$ or $\gg$? Depends on d.f.! Use PTE as a guide.

“Probability to Exceed (PTE)”:

$PTE = 1 - CDF(\chi^2)$ according to chi squared probability density function with $\nu = M - p$ degrees of freedom :

https://en.wikipedia.org/wiki/Chi-squared_distribution



From Wikipedia, the free encyclopedia

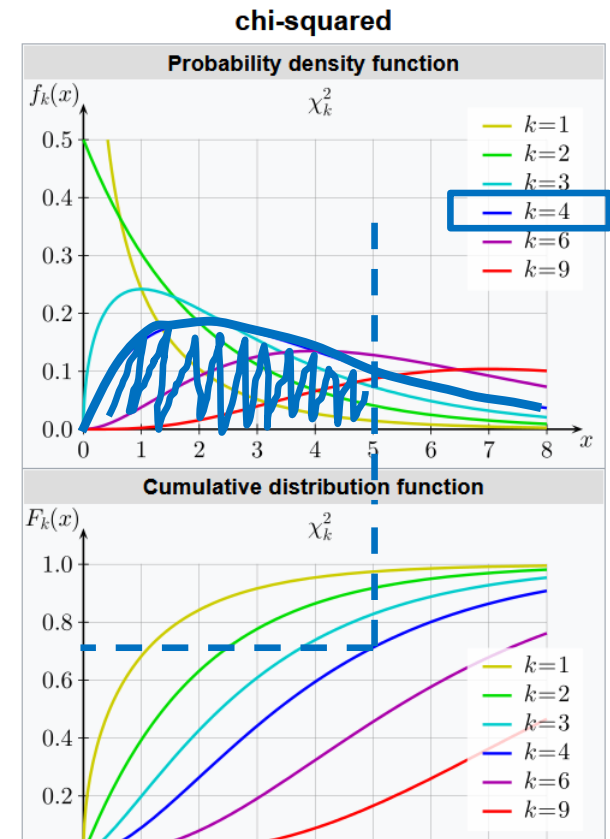
This article is about the mathematics of the chi-squared distribution. For its uses in statistics, see [chi-squared test](#). For the music group, see [Chi2 \(band\)](#).

In [probability theory](#) and [statistics](#), the **chi-squared distribution** (also **chi-square** or χ^2 -distribution) with k [degrees of freedom](#) is the [distribution of a sum of the squares of \$k\$ independent standard normal random variables](#). The chi-squared distribution is a special case of the [gamma distribution](#) and is one of the most widely used [probability distributions](#) in [inferential statistics](#), notably in [hypothesis testing](#) and in construction of [confidence intervals](#).^{[2][3][4][5]} This distribution is sometimes called the **central chi-squared distribution**, a special case of the more general [noncentral chi-squared distribution](#).

The chi-squared distribution is used in the common [chi-squared tests](#) for [goodness of fit](#) of an observed distribution to a theoretical one, the [independence](#) of two criteria of classification of [qualitative data](#), and in confidence interval estimation for a population [standard deviation](#) of a normal distribution from a sample standard deviation. Many other statistical tests also use this distribution, such as [Friedman's analysis of variance by ranks](#).

Definitions [\[edit \]](#)

If $Z_1, \dots, Z_k$ are [independent](#), [standard normal](#) random variables, then the sum of their squares,



Meaning of $\frac{\chi^2}{d.f.}$ and PTE:

$\frac{\chi^2}{d.f.} \ll 1$? PTE larger than 0.99 (or maybe 0.95)

-this experiment was lucky?

-uncertainties of data were overestimated?

(if so, reported uncertainties of fit parameters are suspect)

-different data points were sampled in correlated ways?

$\frac{\chi^2}{d.f.} \gg 1$? PTE smaller than 0.01 (or maybe 0.05)

-this experiment was unlucky?

-uncertainties of data were underestimated?

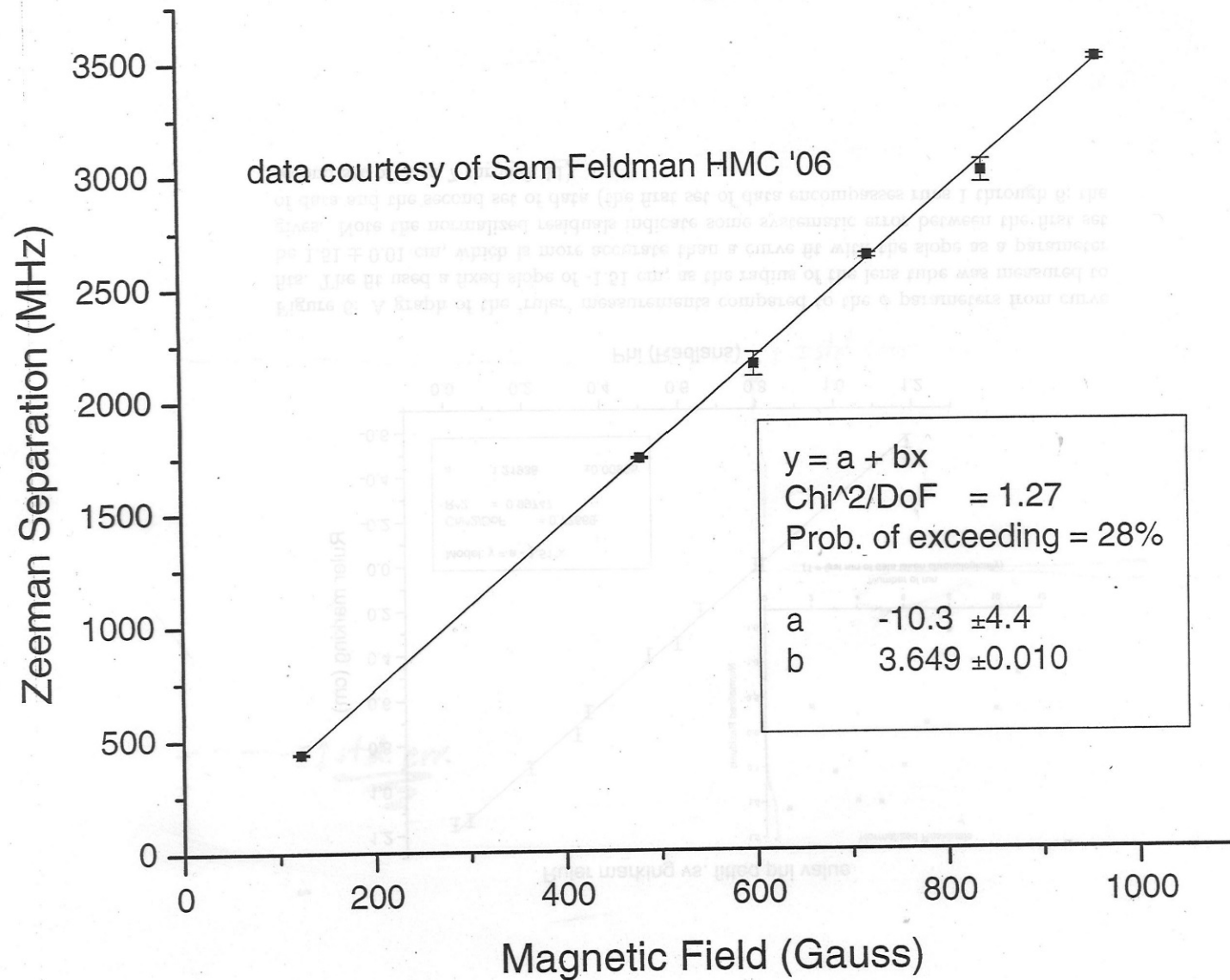
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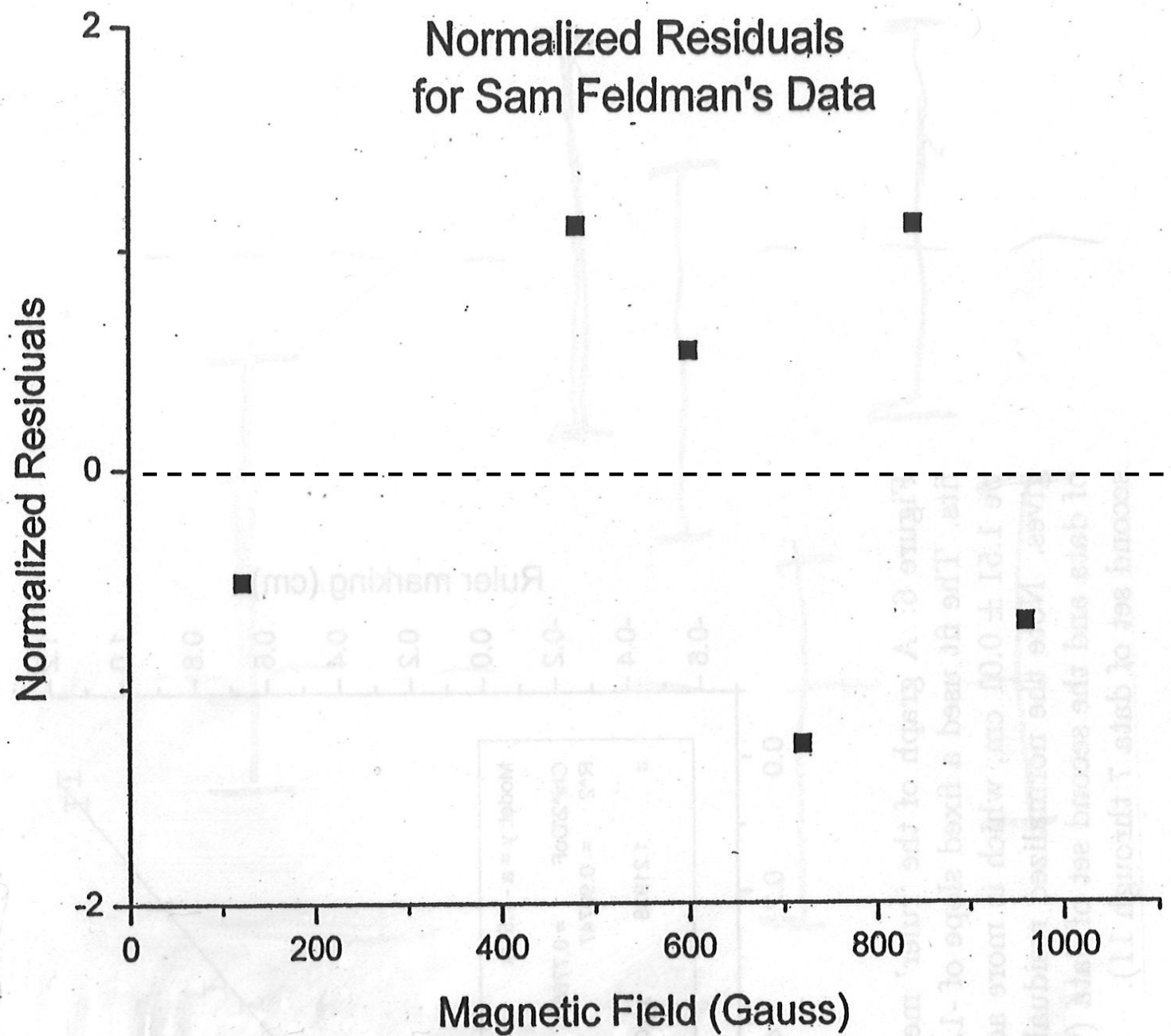
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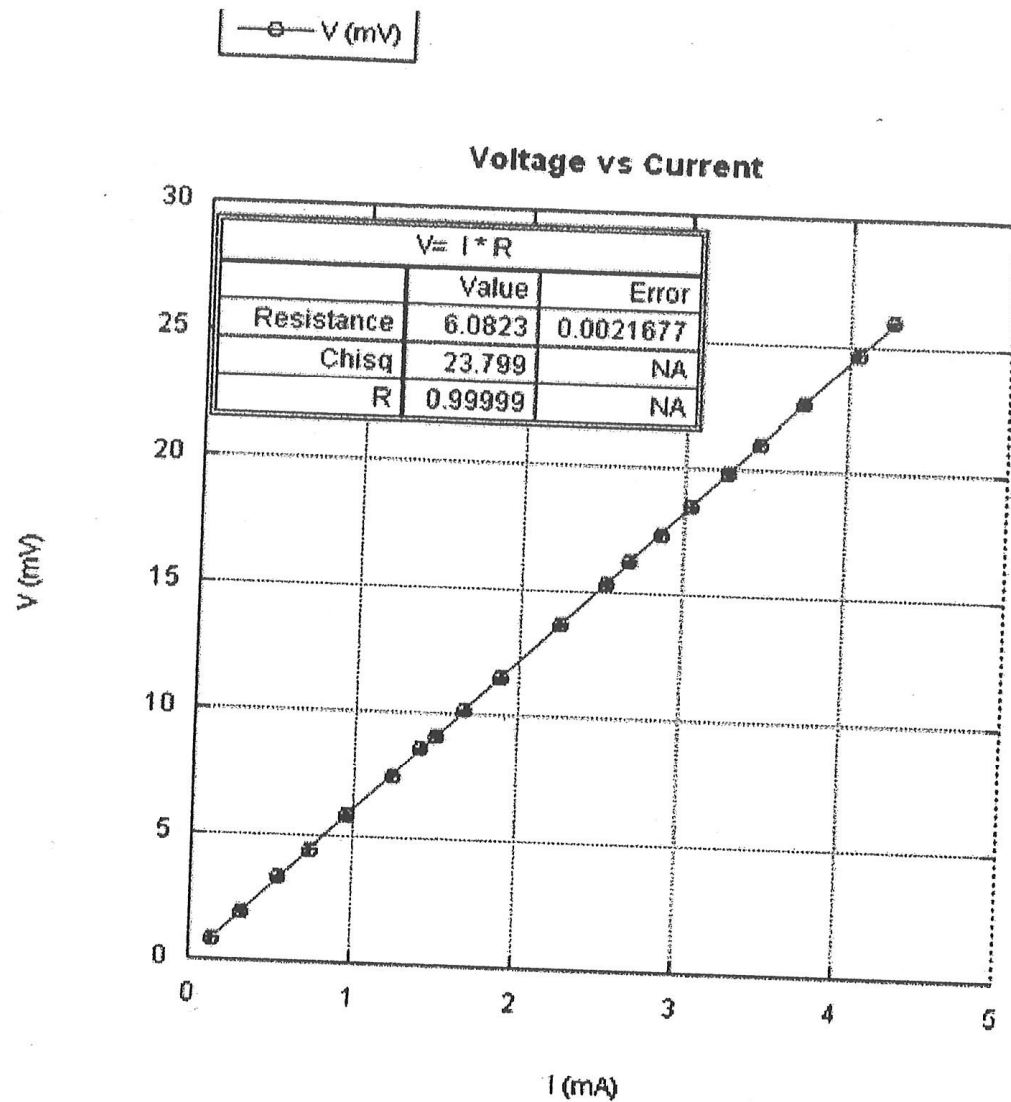
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(wrong or incomplete theory? systematic error not found?)

Zeeman Separation vs. B-Field

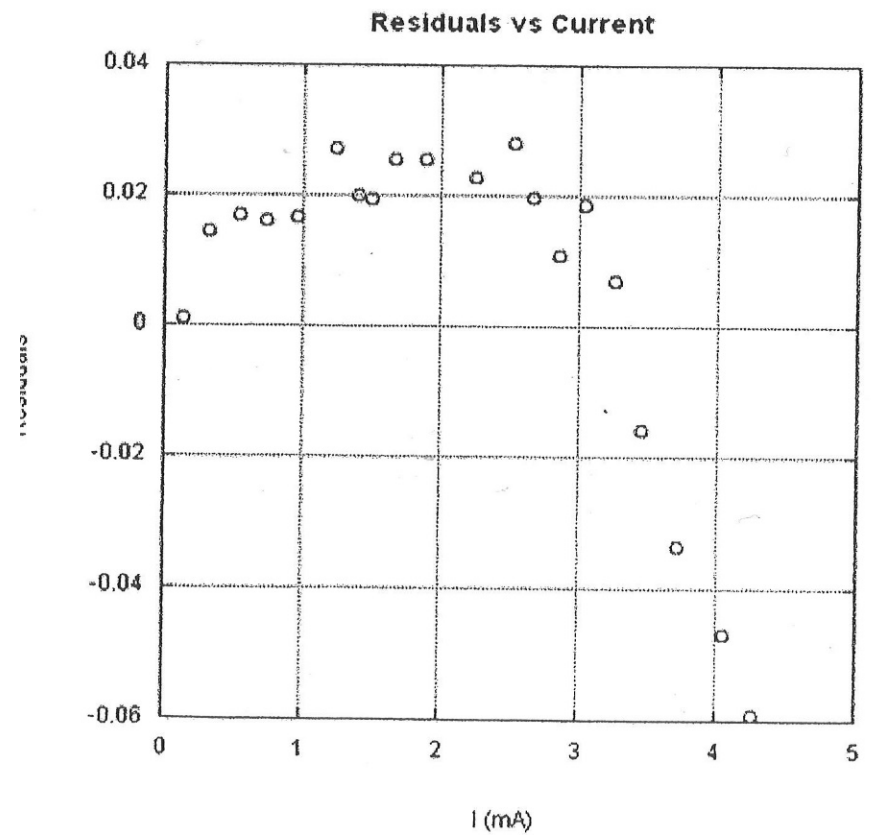




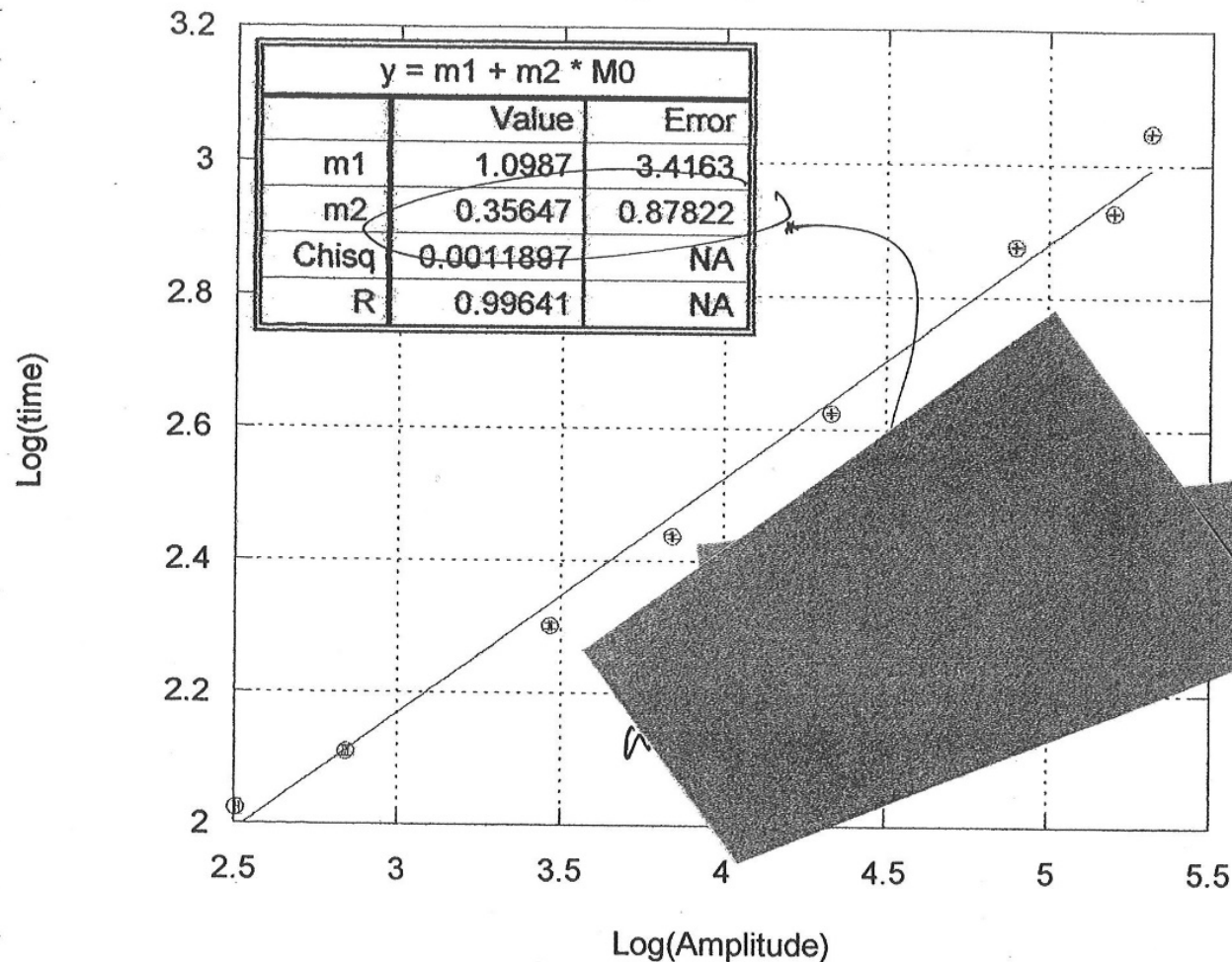


$M=20$ data points

What's shown is the total χ^2 , not $\frac{\chi^2}{d.f.}$.



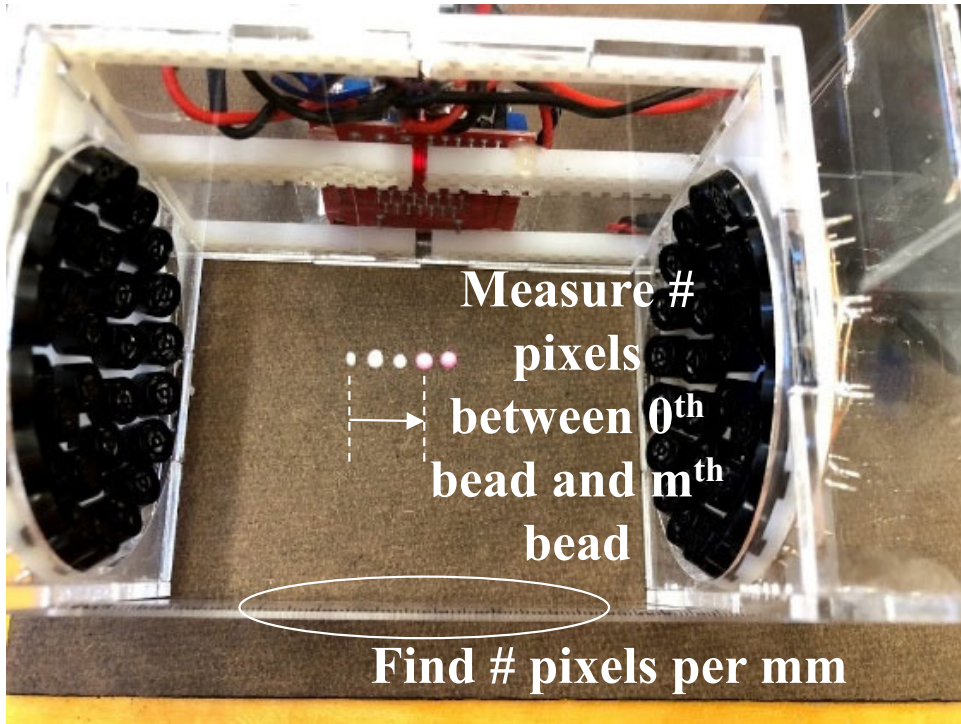
Large Amplitudes



Looked at this example last week:

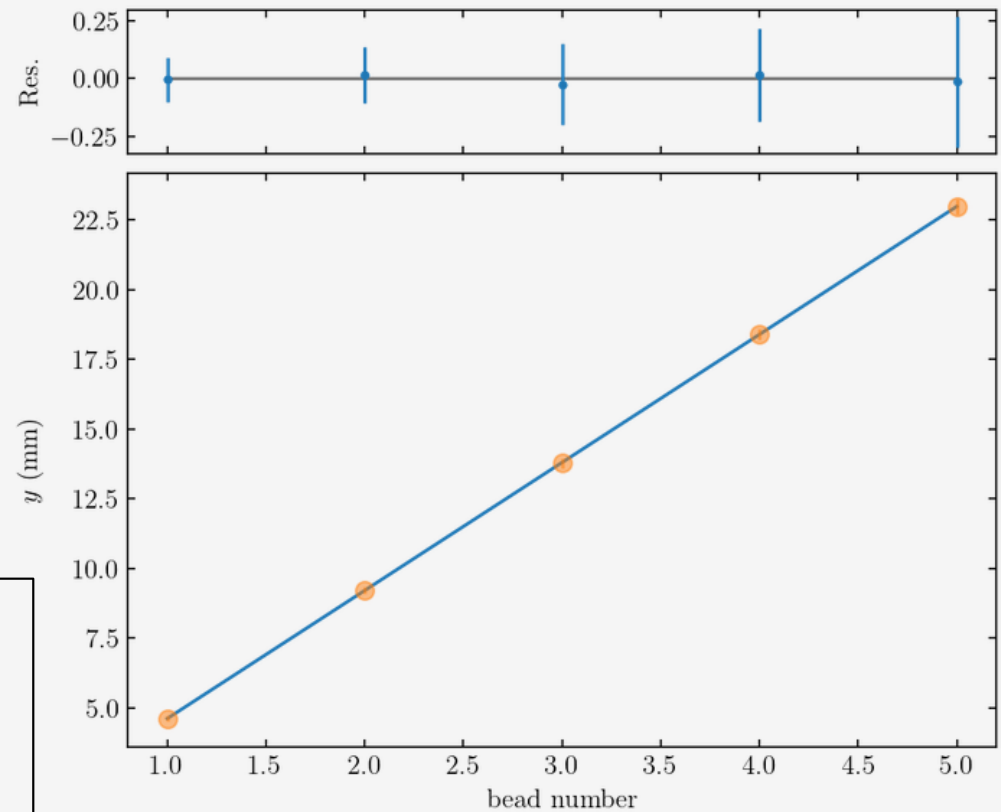
What's shown on the graph is not compatible with the fit outputs listed in the box!

Either the wrong data and/or error bars were plotted, or the wrong data and/or error bars were given to the fit.



**Photo #1 gives 1st trial for each bead.
Reposition ruler and camera so ruler and beads are equal distances from camera,
take next photo, analyze in same way.**

Plot position vs. bead number



$$y = mx + b$$

Param	Value	Unc.
m	4.590	0.054
b	0.04	0.13
$\chi^2_\nu = 0.0154$ for $\nu = 3$		$P_{>} = 99.7\%$

$$y = mx + b$$

Param	Value	Unc.
m	4.590	0.054
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$\chi^2_\nu = 0.0154$ for $\nu = 3$		$P_{>} = 99.7\%$

Depth of Field (m) v. Aperture Width (m)

